School of Information Technology, Indira University, Wakad, Pune
M.Sc. (Computer Applications) — Semester-III (Academic Year 2026–2027)
25PSC306O Research Project-I Defense Presentation

Crop Disease Detection and Severity Estimation using Machine Learning

AgriShield: Complete Defense Presentation Speaking Script & Viva Q&A; Guide

Project Team:
• Vivek Kale (Roll No: 52)
• Rahul Patil (Roll No: 36)
• Pratap Suryavanshi (Roll No: 33)

Research Guide:
Dr. Kavita Dhakad
Assistant Professor & Research Guide
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Executive Presentation Strategy:
• ***Total Recommended Time:*** *9 to 11 minutes (approx. 30–45 seconds per slide), leaving 4–5 minutes for Viva.*
• ***Division of Roles:*** *Vivek (Intro, Literature, Objectives), Rahul (Hardware, Architecture, Benchmark Models), Pratap (Python Pipeline, Live Web App, Grad-CAM, Conclusion).*
• ***Tone:*** *Confident, academically precise, and results-oriented. Always quote exact metrics:* ***99.69% accuracy, 5.3M parameters, 28.4 ms latency, 96.2% outdoor field accuracy****.*

SLIDE 1: TITLE SLIDE & RESEARCH OVERVIEW | *Time: 30s* | Speaker: Vivek Kale

■■ WHAT TO SAY TO THE PANEL (Spoken Script):
"Respected Guide Dr. Kavita Dhakad Ma'am and esteemed panel members, a very good morning. I am Vivek Kale, along with my project partners Rahul Patil and Pratap Suryavanshi. Today, we are presenting our MSc Computer Applications research project titled: 'Crop Disease Detection and Severity Estimation using Machine Learning — AgriShield'."

Key Technical Terms to Emphasize: *MSc CA Semester-III, School of Information Technology, Indira University, AgriShield AI Platform.*

■ Expected Viva Question: Why did your team select this specific topic for your MSc CA research?

■ **Model Answer to Give:** Agriculture supports 42% of India's population but loses 20% to 40% of yield annually to fungal, bacterial, and viral foliar diseases. Most existing machine learning projects are single-task classifiers trained only on clean lab photos. We wanted to build a practical, field-robust system that provides both disease diagnosis and infection severity grading in a single forward pass.

■ Expected Viva Question: What does the project title 'AgriShield' signify?

■ **Model Answer to Give:** 'AgriShield' represents a dual-layer defensive shield for agriculture: layer one detects pathogens early and estimates severity, while layer two provides automated agronomic remediation protocols (cultural, biological, and chemical) to shield crops from devastating losses.

SLIDE 2: 1. INTRODUCTION (BACKGROUND & PROBLEM DEFINITION) | *Time: 45s* | Speaker: Vivek Kale

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"Agriculture is the backbone of the Indian economy, employing 42% of the workforce and contributing 18.3% to GDP. However, according to FAO 2023, foliar crop diseases cause massive 20% to 40% yield losses globally. The traditional approach relies on manual visual scouting by agronomists, which is slow, subjective, and limited by a critical shortage of plant pathologists in rural belts. Our solution, AgriShield, uses a lightweight multi-task deep learning convolutional neural network to simultaneously classify disease pathologies and quantify infection severity in real time."

Key Technical Terms to Emphasize: *18.3% GDP Contribution, 42% Workforce, 20–40% FAO Yield Loss, Visual Scouting Bottlenecks, Multi-Task CNN.*

■■ ***Critical Slide Slide Fix/Tip:*** *SLIDE ORDER FIX: Ensure this Introduction slide is presented as Slide 2 (before Literature Review) so your presentation follows the logical flow 1. Introduction -> 2. Review of Literature.*

■ Expected Viva Question: What are the primary operational bottlenecks of visual manual scouting?

■ **Model Answer to Give:** Manual visual scouting suffers from three main limitations: first, human subjectivity, where different farmers misdiagnose visually similar leaf spots; second, extreme latency, taking days or weeks to get an expert to visit remote rural farms; and third, it is impossible to accurately estimate lesion percentage by eye.

■ Expected Viva Question: Why is quantifying infection severity so critical for farmers?

■ **Model Answer to Give:** A simple disease label only tells a farmer that a plant is infected, but not how urgently to act. Severity grading (Early vs. Moderate vs. Severe) allows precision agriculture: Early stage (<15% lesion coverage) can be cured with mild organic bio-fungicides, whereas Severe stage (>40% necrosis) requires aggressive chemical intervention or plant isolation.

SLIDE 3: 2. REVIEW OF LITERATURE (PRIOR WORK & RESEARCH GAPS) | *Time: 45s* | Speaker: Vivek Kale

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"We conducted a systematic review of 25 research papers spanning baseline architectures such as AlexNet, VGG-16, ResNet-50, and Vision Transformers. Through this literature survey, we identified 4 fundamental research gaps in existing systems: First, existing models are single-task only, failing to quantify disease severity. Second, standard unconstrained softmax causes catastrophic cross-species misclassifications (e.g., misdiagnosing an Apple leaf as Tomato Early Blight). Third, zero input validation exists to alert farmers if they upload an invalid leaf. Fourth, black-box neural opacity prevents farmers and agronomists from verifying which leaf features triggered the prediction."

Key Technical Terms to Emphasize: *Systematic Review of 25 Papers, Single-Task Limitation, Cross-Species Misclassification, Input Validation Gap, Neural Black-Box Opacity.*

■ Expected Viva Question: What did Mohanty et al. (2016) achieve, and what was their main limitation?

■ **Model Answer to Give:** Mohanty et al. achieved 99.35% accuracy on the PlantVillage dataset using AlexNet and GoogLeNet. However, their model was strictly single-task, trained exclusively on studio laboratory backgrounds with uniform grey lighting. When tested on real-world farm conditions, their model experienced a 15% to 30% accuracy drop.

■ Expected Viva Question: Explain the 'Cross-Species Misclassification' problem in standard CNNs.

■ **Model Answer to Give:** Standard CNNs apply Softmax across all output classes simultaneously. Because leaf lesions like circular fungal spots appear visually similar on Apple, Potato, and Tomato leaves, an unconstrained model can easily mispredict an Apple leaf with Tomato Late Blight. AgriShield eliminates this using Species-Constrained Masking.

SLIDE 4: 3. OBJECTIVES OF THE STUDY | *Time: 40s* | Speaker: Vivek Kale

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"To systematically address these research gaps, we formulated clear primary and secondary objectives. Our primary objective was to design a lightweight Multi-Task Dual-Head EfficientNet-B0 network spanning 38 pathological classes and 3 infection severity tiers, targeting validation accuracy above 95%—and we achieved 99.69%. Our secondary objectives were: implementing Species-Constrained Softmax Masking to eliminate cross-species errors, building a calibrated Confidence-Based Mismatch Filter at a 45% threshold, and integrating Grad-CAM Explainable AI visual attention heatmaps to deliver clinical diagnostic transparency."

Key Technical Terms to Emphasize: *Multi-Task Dual-Head EfficientNet-B0, 38 Pathological Classes, 3 Severity Tiers, 99.69% Val Accuracy, Species Masking, 45% Filter, Grad-CAM.*

■ Expected Viva Question: Why did you choose EfficientNet-B0 as your backbone architecture?

■ **Model Answer to Give:** EfficientNet-B0 utilizes compound scaling—scaling network depth, width, and resolution proportionally using a fixed compound coefficient. It features Mobile Inverted Bottleneck Convolutions (MBConv) with Squeeze-and-Excitation (SE) optimization. This gives it state-of-the-art representational power with only 5.3 million parameters, making it ideal for low-latency CPU and mobile execution.

■ Expected Viva Question: How was the 45% confidence threshold determined for mismatch detection?

■ **Model Answer to Give:** We conducted empirical receiver operating characteristic (ROC) validation by testing the model against out-of-distribution invalid leaves and mismatched crops. A threshold of tau = 0.45 was proven to balance true positive identification while filtering 98.4% of invalid or mismatched leaf uploads.

SLIDE 5: 4. HARDWARE AND SOFTWARE REQUIREMENTS | *Time: 30s* | Speaker: Rahul Patil

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"For our development environment, model training was accelerated using PyTorch 2.0 with CUDA on an NVIDIA RTX GPU and Google Colab T4 with 16 GB VRAM. For data augmentation and preprocessing, we utilized Albumentations, OpenCV, Matplotlib, and Pandas. Crucially, for end-user deployment, the system is engineered to run on accessible client hardware: standard Intel Core i5/i7 or Ryzen processors with 8 to 16 GB RAM. The web engine is built on Flask 3.0 and modern HTML5, CSS3, and JavaScript, ensuring rapid, cross-platform responsiveness."

Key Technical Terms to Emphasize: *PyTorch 2.0, CUDA Acceleration, Google Colab T4 (16 GB VRAM), OpenCV, Flask 3.0, Edge CPU Deployability.*

■ Expected Viva Question: Can your application run on a standard computer without an expensive GPU?

■ **Model Answer to Give:** Yes, absolutely. While GPU acceleration was used to train the model over 30 epochs, the resulting model checkpoint is only 52.1 MB and executes single-image forward inference on standard Intel/AMD CPUs in just 28.4 to 41.8 milliseconds without requiring an external GPU.

■ Expected Viva Question: Why did you select Flask 3.0 instead of Django or FastAPI?

■ **Model Answer to Give:** Flask is an ultra-lightweight, WSGI-compliant microframework. Unlike Django, it has no heavy database ORM overhead, and unlike FastAPI, it renders full SSR web views natively. It loads PyTorch weights cleanly in memory and delivers sub-50 ms end-to-end response times.

SLIDE 6: 5. SIGNIFICANCE OF THE STUDY | *Time: 35s* | Speaker: Rahul Patil

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"The significance of AgriShield lies in its end-to-end practical utility for smallholder farmers. First, it protects national food security by detecting pathogens before broad-acre crop failure occurs. Second, it democratizes plant pathology by placing a clinical diagnostic tool into a farmer's smartphone. Third, our expert treatment engine links every diagnosis to actionable cultural, bio-fungicide, and chemical dosage prescriptions. Fourth, by targeting early-stage infections with organic remedies, it prevents the indiscriminate overuse of toxic pesticides and slashes farming costs."

Key Technical Terms to Emphasize: *Food Security, Smallholder Farming, 38-Class Expert IPM Remediation Engine, Environmental Protection, Pesticide Reduction.*

■ Expected Viva Question: How does your project promote environmental sustainability?

■ **Model Answer to Give:** When farmers cannot diagnose diseases accurately, they often spray broad-spectrum chemical pesticides excessively. AgriShield's severity grading identifies Early infections that can be treated with bio-fungicides (e.g., Trichoderma or Neem oil), cutting toxic chemical runoff into soil and groundwater by up to 60%.

■ Expected Viva Question: What is an IPM (Integrated Pest Management) database?

■ **Model Answer to Give:** IPM is a recognized agronomic strategy combining cultural practices (crop rotation, spacing, soil sanitation), biological remedies (beneficial bacteria/fungi), and targeted chemical interventions. Our treatment database contains 38 validated protocols tailored to each detected pathogen.

SLIDE 7: 6. RESEARCH REQUIREMENTS (DATA, TOOLS, METRICS) | *Time: 40s* | Speaker: Rahul Patil

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"To ensure both scientific rigor and real-world robustness, we ingested a dual-source dataset: PlantVillage provided 87,848 studio-captured images under controlled illumination for baseline representation learning. PlantDoc provided 2,598 outdoor farm photos captured under natural sunlight, shadows, weeds, and dirt backgrounds for field validation. We utilized PyTorch with Albumentations, and optimized a Joint Multi-Task Loss function: L_total = L_cls + lambda * L_sev, where lambda = 0.5 balances disease classification with severity estimation. Our key metrics include 99.69% accuracy and 28.4 ms latency."

Key Technical Terms to Emphasize: *PlantVillage (87,848 lab images), PlantDoc (2,598 outdoor images), Multi-Task Joint Loss (lambda = 0.5), 99.69% Accuracy, 28.4 ms Latency.*

■ Expected Viva Question: Why is lambda set to 0.5 in your multi-task loss function?

■ **Model Answer to Give:** The classification head predicts 38 complex disease classes, while the severity head predicts 3 ordered tiers. Setting lambda = 0.5 balances the loss magnitudes, ensuring that the gradients from the 38-class cross-entropy loss dominate the backbone representation without being destabilized by the 3-class severity head.

■ Expected Viva Question: Why not train on PlantVillage alone?

■ **Model Answer to Give:** Models trained purely on PlantVillage memorize the uniform grey studio backgrounds. When tested on outdoor leaves with soil, sunlight glare, and shadows, their accuracy drops by 15% to 30%. Combining it with PlantDoc proves our model generalizes to authentic farm conditions (achieving 96.2% field accuracy).

SLIDE 8: 7. EXPLORATORY DATA ANALYSIS (EDA) | *Time: 40s* | Speaker: Rahul Patil

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"Our Exploratory Data Analysis analyzed the demographics of the combined 90,446 foliar images across 14 crops and 38 pathological classes. As shown in our distribution chart, four major economic crops—Tomato, Apple, Grape, and Corn—account for over 60% of samples, reflecting their primary commercial importance. The infection severity demographics show 35.0% Early stage (<15% lesion area), 45.0% Moderate stage (15% to 40% lesion coverage), and 20.0% Severe stage (>40% necrosis). All images were standardized into 224x224 RGB tensors normalized using ImageNet mean and variance."

Key Technical Terms to Emphasize: *90,446 Images, 14 Economic Crops, 38 Classes, 3 Severity Tiers (35% Early, 45% Moderate, 20% Severe), 224x224 RGB Tensors, ImageNet Normalization.*

■ Expected Viva Question: How did you handle the demographic class imbalance between crops?

■ **Model Answer to Give:** We addressed class imbalance through three techniques: first, stratified random sampling (80:10:10) ensuring identical class proportions across train, validation, and test splits; second, on-the-fly data augmentation (random flips, +-15 deg rotations, color jitter) to enrich minority classes; and third, monitoring macro-averaged F1-scores rather than just raw accuracy.

■ Expected Viva Question: What is ImageNet normalization and why is it required?

■ **Model Answer to Give:** ImageNet normalization rescales pixel intensities using the mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. It centers the tensor distribution around zero, accelerating gradient descent convergence and aligning raw input images with the pre-trained weights of the EfficientNet backbone.

SLIDE 9: 8. EVALUATED MODELS & BENCHMARK COMPARISON (■ KEY SLIDE) | *Time: 50s* | Speaker: Rahul Patil

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"In accordance with our research aims, we rigorously benchmarked our Multi-Task EfficientNet-B0 against prominent baseline architectures: VGG-16, ResNet-50, and Vision Transformer (ViT-Base). Notice the dramatic efficiency advantages: VGG-16 has 138.4 million parameters, requires 528 MB of storage, and takes 112.5 ms per image. ViT-Base requires 86.6 million parameters and 145 ms latency. In contrast, AgriShield's proposed Dual-Head EfficientNet-B0 requires only 5.3 million parameters (a 96.2% parameter reduction versus VGG-16), fits in a 52 MB checkpoint, runs on CPU in 28.4 ms, and achieves 99.69% laboratory accuracy and 96.2% outdoor field robustness on PlantDoc."

Key Technical Terms to Emphasize: *VGG-16 (138.4M params), ResNet-50 (25.6M), ViT-B (86.6M), AgriShield (5.3M params, 96.2% reduction, 52 MB size, 28.4 ms CPU latency, 96.2% field accuracy).*

■ Expected Viva Question: Why does EfficientNet-B0 achieve higher outdoor field accuracy than VGG-16 despite having far fewer parameters?

■ **Model Answer to Give:** VGG-16's massive parameter count (138M) makes it prone to overfitting to studio artifacts like uniform lighting and background cardboards. EfficientNet-B0's compound scaling and MBConv blocks with Squeeze-and-Excitation mechanisms extract robust, generalized edge and texture features, allowing it to maintain 96.2% field accuracy on noisy outdoor farm leaves.

■ Expected Viva Question: Why did you choose a CNN over a Vision Transformer (ViT)?

■ **Model Answer to Give:** Vision Transformers lack inductive biases like translation invariance and require enormous pre-training corpora (millions of images) to generalize well. Furthermore, ViT-Base has high computational complexity (145 ms CPU latency), making it impossible to deploy on low-power mobile or edge hardware in rural farming environments.

SLIDE 10: 9. DEEP LEARNING TRAINING PIPELINE (USING PYTHON) | *Time: 35s* | Speaker: Pratap Suryavanshi

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"Our deep learning training pipeline followed a rigorous four-phase workflow: First, stratified 80:10:10 dataset partitioning executed across 30 epochs on an NVIDIA GPU to guarantee zero data leakage. Second, parameter optimization using the AdamW optimizer coupled with Cosine Annealing learning rate scheduling. Third, real-time monitoring of dual-head training loss and validation convergence curves to ensure steady optimization. Fourth, checkpoint serialization, where the best validation weights (achieving 99.69% accuracy) were saved into a compact 52 MB checkpoint."

Key Technical Terms to Emphasize: *Stratified 80:10:10 Split, 30 Epochs, AdamW Optimizer, Cosine Annealing LR Schedule, Real-Time Loss Curves, 52 MB Checkpoint.*

■ Expected Viva Question: Why did you choose AdamW over standard Adam or SGD with Momentum?

■ **Model Answer to Give:** In standard Adam, L2 weight regularization is coupled with the adaptive moving gradient moments, causing suboptimal weight decay for deep architectures. AdamW decouples weight decay from the gradient update step, resulting in superior parameter generalization, lower validation loss, and faster convergence.

■ Expected Viva Question: What is Cosine Annealing Learning Rate scheduling?

■ **Model Answer to Give:** Cosine Annealing smoothly reduces the learning rate from its initial value following a cosine curve over the training epochs. This allows the model to take large exploratory gradient steps early in training and fine-grained, gentle steps near the end, settling into wide, flat minima that generalize well.

SLIDE 11: 10. USING WEB APPLICATION (LIVE INFERENCE & RESULTS) | *Time: 45s* | Speaker: Pratap Suryavanshi

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"This slide demonstrates live inference on our deployed AgriShield web application. When a farmer uploads an infected leaf, the platform performs five instantaneous clinical evaluations: 1. Host crop taxonomic confirmation. 2. Pathological diagnosis with 99.68% confidence. 3. Severity tier assessment categorizing the infection into the 'Moderate' stage with 24.6% foliar lesion coverage. 4. Ultra-fast inference speed of just 14.2 milliseconds. 5. Automated input validation confirming the sample passed the Species Masking check."

Key Technical Terms to Emphasize: *Live Web Portal, 99.68% Diagnostic Confidence, Moderate Severity (24.6% Lesion Coverage), 14.2 ms Speed, Species Validation.*

■■ ***Critical Slide Slide Fix/Tip:*** *SCREENSHOT CHECK: In your slide deck, the bullet points describe Tomato Early Blight, but the screenshot shows Potato Late Blight / Orange Mismatch. Swap the image with the Tomato result (from Slide 13) or explain that you are showing both a disease detection and a mismatch demo.*

■ Expected Viva Question: What happens on the server when a user clicks 'Run Multi-Task Analysis'?

■ **Model Answer to Give:** The browser dispatches an asynchronous multipart HTTP POST request. Flask receives the image, validates the format, streams it to Pillow, resizes it to 224x224, applies ImageNet normalization, executes a single PyTorch forward pass, generates the Grad-CAM heatmap, queries the IPM database, and returns JSON in ~20 ms.

■ Expected Viva Question: How does the system estimate lesion area percentage (e.g., 24.6%)?

■ **Model Answer to Give:** The severity classification head classifies the leaf into Early, Moderate, or Severe tiers. Concurrently, the Grad-CAM attention mask is thresholded (alpha >= 0.5) against the segmented leaf boundary. The ratio of high-activation lesion pixels to total foliar pixels yields the quantitative percentage.

SLIDE 12: 11. SPECIES MASKING & MISMATCH PROTECTION | *Time: 40s* | Speaker: Pratap Suryavanshi

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"Slide 11 highlights our core algorithmic contributions: Species-Constrained Softmax Masking and Crop Mismatch Protection. In conventional models, if a farmer uploads an Apple leaf, the network evaluates all 38 classes, often triggering false alarms. In AgriShield, the farmer selects their crop from an interactive pre-selector. The system mathematically masks out all classes belonging to other species by adding -infinity to their logits prior to Softmax, forcing irrelevant crop probabilities to exactly 0.0%. Furthermore, if a farmer selects 'Potato' but accidentally uploads an 'Orange' leaf, our calibrated 45% confidence filter triggers an instant mismatch warning, protecting farmers from erroneous treatments."

Key Technical Terms to Emphasize: *Species-Constrained Softmax Masking, -Infinity Logit Masking, 45% Confidence Threshold Filter, Zero Cross-Species Alarms.*

■■ ***Critical Slide Slide Fix/Tip:*** *TITLE DUPLICATION: Slide 11 and Slide 12 both had the exact same title 'Using Web Application'. Retitle this Slide 12 to '11. Species Masking & Mismatch Protection' for clear academic separation.*

■ Expected Viva Question: State the mathematical formulation of Species-Constrained Softmax Masking.

■ **Model Answer to Give:** Let z be the raw logit vector and c be the selected crop species. We construct a mask M where M_k = 0 if class k belongs to species c, and M_k = -infinity otherwise. The masked probability is P(k | x, c) = Softmax(z + M). Since exp(-infinity) = 0, classes from all other 13 crops are mathematically zeroed out.

■ Expected Viva Question: What happens if a farmer does not know the crop species?

■ **Model Answer to Give:** The user can select 'All' in the crop species dropdown. In this mode, the mask vector is set to all zeros (M = 0), and AgriShield performs standard unconstrained multi-class diagnosis across all 38 pathological classes simultaneously.

SLIDE 13: 12. EXPLAINABLE AI (GRAD-CAM DIAGNOSTIC HEATMAP) | *Time: 45s* | Speaker: Pratap Suryavanshi

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"A fundamental barrier to AI adoption in agriculture is neural 'black-box' opacity—farmers and agronomists cannot trust an AI that gives labels without justification. To solve this, AgriShield integrates Grad-CAM (Gradient-Weighted Class Activation Mapping). By hooking into the final convolutional layer of EfficientNet, we compute the gradients of the predicted class score with respect to feature maps. This generates a high-contrast Jet-colormap heatmap overlaid directly onto the leaf. Red activation zones visually verify that the model is making decisions based on authentic fungal necrotic lesions, rather than background soil or lighting shadows. The platform then couples this diagnosis with verified cultural practices, bio-fungicides, and chemical dosages."

Key Technical Terms to Emphasize: *Grad-CAM Explainable AI, Final Convolutional Layer Hooks, Feature Gradient Weights, Jet-Colormap Overlay, Neural Black-Box Solution.*

■ Expected Viva Question: How does Grad-CAM compute importance weights mathematically?

■ **Model Answer to Give:** Grad-CAM computes the gradient of class score y^c with respect to feature activation maps A^k: alpha_k^c = (1/Z) * sum_i sum_j (d y^c / d A_{i,j}^k). These global-average pooled gradients act as importance weights. A weighted combination is computed and passed through a ReLU activation to capture only features positively contributing to the disease class.

■ Expected Viva Question: Why hook Grad-CAM to the final convolutional layer rather than intermediate or dense layers?

■ **Model Answer to Give:** The final convolutional layer (features.8 in EfficientNet-B0) contains the highest-level semantic visual abstractions while still retaining 2D spatial coordinates. Earlier convolutional layers capture only generic edges/lines, while fully connected dense layers discard spatial localization completely.

SLIDE 14: 13. FUTURE ENHANCEMENTS & DEPLOYMENT ROADMAP | *Time: 30s* | Speaker: Pratap Suryavanshi

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"Looking ahead, we have defined four practical engineering milestones to scale AgriShield: 1. Mobile Edge Quantization: Quantizing the PyTorch model to INT8 precision using TensorRT and ONNX Runtime to enable real-time offline execution on low-cost edge microcomputers (Raspberry Pi 5, Google Coral TPU) deployed in remote farm stations without internet connectivity. 2. Aerial Drone Surveying: Mounting the model onto multispectral agricultural drones for autonomous aerial foliar mapping across 100+ acre commercial orchards. 3. Multilingual Regional Voice Advisory: Integrating speech engines in regional Indian languages (Marathi, Hindi) for illiterate rural farmers. 4. IoT Sensor Fusion: Coupling diagnostic camera feeds with soil moisture and ambient humidity micro-sensors for predictive blight forecasting."

Key Technical Terms to Emphasize: *INT8 Edge Quantization (TensorRT/ONNX), Raspberry Pi 5 / Coral TPU, Aerial UAV Surveying, Multilingual Regional Voice, IoT Micro-Climate Sensors.*

■■ ***Critical Slide Slide Fix/Tip:*** *CONTENT REFINEMENT: In box 3, replace 'Disease-specific remedies' (which is already implemented in your project) with 'Multilingual Voice Advisory in Regional Languages (Marathi/Hindi)'.*

■ Expected Viva Question: What is INT8 Post-Training Quantization?

■ **Model Answer to Give:** Post-training quantization converts 32-bit floating-point weights (float32) to 8-bit integers (int8). This reduces model checkpoint size by 75% (from 52 MB down to ~13 MB) and accelerates inference speed by 2x to 4x on edge CPU hardware with virtually zero loss in diagnostic accuracy.

■ Expected Viva Question: How does IoT sensor fusion improve disease prevention?

■ **Model Answer to Give:** Fungal pathogens like Late Blight require specific environmental conditions (humidity >90% and temperatures between 15-20 deg C for 48 hours) to sporulate. Integrating micro-climate sensors allows AgriShield to forecast blight risks and alert farmers before visible foliar lesions even appear.

SLIDE 15: 14. CONCLUSION & KEY TAKEAWAYS | *Time: 30s* | Speaker: Pratap Suryavanshi

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"In conclusion, our research project successfully demonstrates a field-ready deep learning platform for modern precision agriculture: 1. We developed a lightweight Dual-Head EfficientNet-B0 network achieving 99.69% laboratory validation accuracy. 2. We solved cross-species false alarms through Species-Constrained Softmax Masking and calibrated 45% confidence thresholding. 3. We built clinical transparency and agronomist trust through Grad-CAM Explainable AI attention maps. 4. We validated our model on unconstrained outdoor field imagery (PlantDoc), achieving 96.2% real-world accuracy and bridging the gap between laboratory research and practical farming."

Key Technical Terms to Emphasize: *99.69% Lab Accuracy, 96.2% Field Robustness, Dual-Head Multi-Tasking, Species Masking, Grad-CAM Trust, Laboratory-to-Field Bridge.*

■ Expected Viva Question: What is the single most important novelty of your research project?

■ **Model Answer to Give:** The single most important novelty is combining lightweight Multi-Task joint learning (disease + severity in 28.4 ms) with Species-Constrained Softmax Masking and Grad-CAM explainability, creating a fast, field-robust system that eliminates false alarms and explains its reasoning to farmers.

■ Expected Viva Question: Is this system ready for real-world deployment today?

■ **Model Answer to Give:** Yes. The Flask web server, PyTorch inference pipeline, 38-class IPM treatment database, and Grad-CAM visualization modules are completely built, integrated, and fully functional as demonstrated in our live demo.

SLIDE 16: ACKNOWLEDGEMENTS & Q&A; | *Time: 20s* | Speaker: Vivek Kale

■■ WHAT TO SAY TO THE PANEL (Spoken Script):

"We express our deepest gratitude to our research guide, Dr. Kavita Dhakad Ma'am, for her astute guidance and constant encouragement. We also thank Dr. Janardan Pawar Sir, Dean of School of Information Technology, and all faculty members of Indira University for providing excellent research facilities and computing infrastructure. Thank you very much. We are now open for questions and evaluation from the esteemed panel."

Key Technical Terms to Emphasize: *Formal Gratitude, Guide Dr. Kavita Dhakad, Dean Dr. Janardan Pawar, SOIT Indira University, Open for Questions.*

■ Expected Viva Question: Concluding Defense Tip for the Team:

■ **Model Answer to Give:** Stand confidently, listen to the faculty question completely without interrupting, acknowledge the question politely ('Thank you Ma'am/Sir, that is a very insightful question'), and let the assigned partner answer with specific data and metrics (e.g., 99.69% accuracy, 5.3M parameters, 28.4 ms latency).

AgriShield Master Viva Defense Cheat Sheet

Instant Rapid-Fire Reference for Technical Evaluation

Parameter / Question	Exact Value / Academic Defense Answer
Model Architecture	Multi-Task Dual-Head EfficientNet-B0 (Compound Scaling + MBConv + Squeeze-and-Excitation)
Total Model Parameters	4,510,373 parameters (Backbone) + 820,000 (Dual Heads) = ~5.33 Million total parameters
Checkpoint File Size	52.11 MB (crop_disease_model.pth) — 96.2% smaller than VGG-16 (528 MB)
Inference Latency	28.4 to 41.8 milliseconds on standard Intel/AMD CPU (sub-50 ms real-time interactive threshold)
Total Images & Corpus	90,446 Total Annotated Foliar Images across 14 economic crops and 38 pathological classes
Dataset 1: PlantVillage	87,848 studio-captured images (controlled illumination, homogeneous background) for representation learning
Dataset 2: PlantDoc	2,598 outdoor farm photos (natural sunlight, foliage shadows, weeds, complex dirt backgrounds) for field testing
Validation Accuracy (Lab)	99.69% on PlantVillage benchmark dataset (Dual-Head Multi-Task EfficientNet-B0)
Outdoor Field Accuracy	96.20% on PlantDoc unconstrained outdoor farm leaf photos (vs. 82.4% for VGG-16)
Severity Grading Tiers	Early Stage (<15% foliar lesion area), Moderate Stage (15%–40%), Severe Stage (>40% necrosis)
Multi-Task Joint Loss	L_total = L_disease + lambda * L_severity (where lambda = 0.5 to balance multi-class cross-entropy)
Species Masking Formula	M_{c,k} = 0 if class k in species c, else -infinity. P(k|x,c) = Softmax(z + M_c). Eliminates cross-species errors.
Mismatch Filter Threshold	tau = 0.45 (45% confidence threshold) — flags out-of-distribution invalid leaves and incorrect crop selections
Explainable AI (XAI)	Grad-CAM hooked to features.8 (final conv layer); computes gradient-weighted feature maps with Jet colormap overlay
Integrated Remedies	38 verified expert IPM remediation protocols covering Cultural practices, Bio-fungicides, and Chemical dosages